Lab Experiment-II: Visualization and quantification of free and forced vortices

Objectives

(i) Generate forced and free vortices and visualize the associated free surfaces.

(ii) Measure the location of the free surface to compare with theoretical estimates.

The Essential Background

1. What are free and forced vortices? Briefly discuss a few applications/occurrences of such vortices.

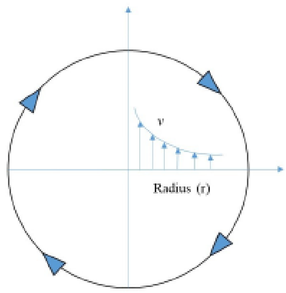
2. Can you quantify velocity profiles and the associated free surfaces for free and forced vortices?

Task 2.1: In the report for Lab Exp.-III , concise answers to all the above questions have to be submitted.

(a) Free Vortex

(b) Forced vortex

(c) Free vortex velocity profile



(d) Forced vortex velocity profile

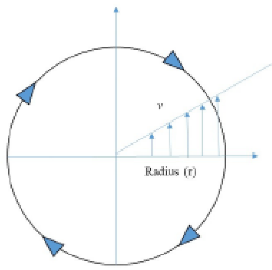


Figure 2: Free and forced vortices.

About the experiment

A typical experimental set up for the free and forced vortices have been shown in figures 2(a) and (b) respectively. The set up for the free vortex simply contains a cylindrical container with a small orifice at its base, through which the liquid in the container may be drained. In order to sustain the vortex at steady state, liquid may be supplied to the container at the same rate as the liquid outflow. Once a steady vortex has been established, the free surface position w.r.t to a reference plane may be measured using a scale.

The forced vortex on the other hand contains a similar cylindrical container (without any orifice at the bottom) resting on a platform that can be rotated mechanically. This causes the liquid inside the container to undergo rigid body rotation, which in turn causes the pressure to vary across the container, leading to a parabolic shape of the free surface. Both of these experiments will be explained during the class hours. The students are required to perform these experiments and take multiple measurements of the free surface. They may take help from the TAs and the lab staff.

Task 2.2:

(a) Measure shape of the free surface for a free vortex at steady state for various orifice sizes (if available).

(b) Measure the position of the free surface for the forced vortex for multiple angular velocities of the platform.

(c) Theoretically compute the free surface shapes for free and forced vortices. Jot down all the assumptions necessary. Now compare your computed free surface shapes to the experimental observed ones. You should quantify experimental uncertainties and errors.

(d) In case there are mismatches between observed and the calculated forces, can you specify some reasons which may cause these mismatches?

Tasks 2.1 and 2.2 should be included in the report for this experiment.